National University of Singapore

MA3236—Non-Linear Programming

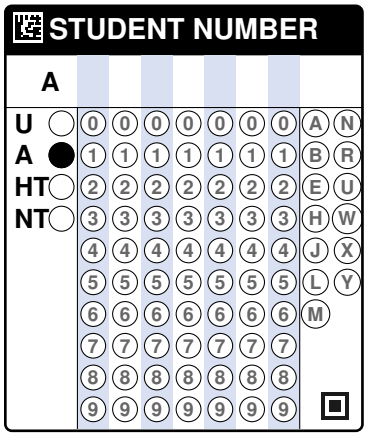
Final Exam

Semester 1, 2025/2026

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS

1. Write down your **student number** on the right and using ink or pencil, shade the corresponding circle **completely** in the grid for each digit or letter. DO NOT WRITE YOUR NAME!
2. The assessment paper contains **FIVE (5) questions** and comprises **SIXTEEN (16) pages** including this cover page.
3. The time allowed for solving this test is **2 hours**.
4. Students are required to answer ALL questions.
5. Students should write the answers for each question on the given spaces.
6. This is a CLOSED BOOK assessment.
7. Students are allowed to use one A4-sized helpsheet.
8. Calculators are allowed. However, students should lay out systematically the various steps in the calculations.



For Examiner's Use Only

Qns	Marks
Q1	/ 18
Q2	/ 25
Q3	/ 22
Q4	/ 12
Q5	/ 23
Total	/100

1. In parts (a)–(c), consider the unconstrained minimization problem:

$$\min_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) := \frac{1}{2}\mathbf{x}^T A\mathbf{x} + \mathbf{c}^T\mathbf{x}.$$

(a) **[5 points]** Suppose $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Given the initial point $\mathbf{x}^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, run two iterations of the steepest descent method with step size 1 to obtain $\mathbf{x}^{(2)}$. Does the steepest descent method converge to the minimizer within two iterations?

Solution. $\nabla f(\mathbf{x}) = A\mathbf{x} + \mathbf{c}$. At $\mathbf{x}^{(0)} = \mathbf{0}$, we have $\nabla f(\mathbf{x}^{(0)}) = \mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

$$\begin{aligned} \mathbf{x}^{(1)} &= \mathbf{x}^{(0)} - \nabla f(\mathbf{x}^{(0)}) = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \\ \mathbf{x}^{(2)} &= \mathbf{x}^{(1)} - \nabla f(\mathbf{x}^{(1)}) = \begin{pmatrix} -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ \nabla f(\mathbf{x}^{(2)}) &= [1;1] \neq \mathbf{0} \end{aligned}$$

The steepest descent method does not converge in two iterations.

(b) **[5 points]** Suppose $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Given the initial point $\mathbf{x}^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, run 2 iterations of the Newton method with step size 1 to obtain $\mathbf{x}^{(2)}$. Does Newton method converge to the minimizer within two iterations?

Solution.

$$\begin{aligned} \mathbf{x}^{(1)} &= \mathbf{x}^{(0)} - H_f^{-1}\nabla f(\mathbf{x}^{(0)}) = -\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \\ \mathbf{x}^{(2)} &= \mathbf{x}^{(1)} - H_f^{-1}\nabla f(\mathbf{x}^{(1)}) = \mathbf{x}^{(1)} - A^{-1}(A\mathbf{x}^{(1)} + \mathbf{c}) - A^{-1}\mathbf{c} = \mathbf{x}^{(1)} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \\ \nabla f(\mathbf{x}^{(2)}) &= \mathbf{0} \end{aligned}$$

Yes, in fact, Newton method terminates in just 1 iteration.

(c) **[8 points]** Suppose $\mathbf{p} \in \mathbb{R}^n$ is a descent direction, $\bar{\mathbf{x}} \in \mathbb{R}^n$ is a given iterate, and $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite. Consider the line search problem

$$\alpha^* = \arg\min_{\alpha\geq 0} f(\bar{\mathbf{x}} + \alpha\mathbf{p}).$$

(i) Write down α^* explicitly in terms of $\mathbf{p}$, A, and $\nabla f(\bar{\mathbf{x}})$.

(ii) Suppose $\sigma \in (0, 0.5)$. Does the optimal step size α^* always satisfy the Armijo condition below?

$$f(\bar{\mathbf{x}} + \alpha^*\mathbf{p}) \leq f(\bar{\mathbf{x}}) + \alpha^*\sigma\nabla f(\bar{\mathbf{x}})^T\mathbf{p}$$

Justify your answer.

Solution. α^* is the stationary point of $f(\bar{\mathbf{x}} + \alpha\mathbf{p})$:

$$\begin{aligned}\alpha^* &= \{\alpha : \mathbf{p}^T\nabla f(\bar{\mathbf{x}} + \alpha\mathbf{p}) = \mathbf{p}^T(A(\bar{\mathbf{x}} + \alpha\mathbf{p}) + \mathbf{c}) = \mathbf{0}\} \\ &= -\frac{\mathbf{p}^T(A\bar{\mathbf{x}} + \mathbf{c})}{\mathbf{p}^TA\mathbf{p}} = -\frac{\mathbf{p}^T\nabla f(\bar{\mathbf{x}})}{\mathbf{p}^TA\mathbf{p}} \geq 0 \quad \text{since } \mathbf{p}^T\nabla f(\bar{\mathbf{x}}) \leq 0\end{aligned}$$

Then the Armijo condition gives:

$$\begin{aligned}& f(\bar{\mathbf{x}}) + \alpha^*\sigma\nabla f(\bar{\mathbf{x}})^T\mathbf{p} - f(\bar{\mathbf{x}} + \alpha^*\mathbf{p}) \\ = \;& \frac{1}{2}\bar{\mathbf{x}}^TA\bar{\mathbf{x}} + \mathbf{c}^T\bar{\mathbf{x}} - \sigma\frac{(\mathbf{p}^T\nabla f(\mathbf{x}))^2}{\mathbf{p}^TA\mathbf{p}} - \frac{1}{2}(\bar{\mathbf{x}} + \alpha^*\mathbf{p})^TA(\bar{\mathbf{x}} + \alpha^*\mathbf{p}) - \mathbf{c}^T(\bar{\mathbf{x}} + \alpha^*\mathbf{p}) \\ = \;& -\sigma\frac{(\mathbf{p}^T\nabla f(\mathbf{x}))^2}{\mathbf{p}^TA\mathbf{p}} - \alpha^*\bar{\mathbf{x}}^TA\mathbf{p} - \frac{(\alpha^*)^2}{2}\mathbf{p}^TA\mathbf{p} - \alpha^*\mathbf{c}^T\mathbf{p} \\ = \;& -\sigma\frac{(\mathbf{p}^T\nabla f(\mathbf{x}))^2}{\mathbf{p}^TA\mathbf{p}} - \frac{(\mathbf{p}^T\nabla f(\bar{\mathbf{x}}))^2}{2\mathbf{p}^TA\mathbf{p}} - \alpha^*(A\bar{\mathbf{x}} + \mathbf{c})^T\mathbf{p} \\ = \;& -\sigma\frac{(\mathbf{p}^T\nabla f(\mathbf{x}))^2}{\mathbf{p}^TA\mathbf{p}} - \frac{(\mathbf{p}^T\nabla f(\bar{\mathbf{x}}))^2}{2\mathbf{p}^TA\mathbf{p}} + \frac{(\mathbf{p}^T\nabla f(\bar{\mathbf{x}}))^2}{\mathbf{p}^TA\mathbf{p}} \\ = \;& (\frac{1}{2} - \sigma)\frac{(\mathbf{p}^T\nabla f(\mathbf{x}))^2}{\mathbf{p}^TA\mathbf{p}} \geq 0\end{aligned}$$

since $\sigma < 1/2$, $\mathbf{p}^TA\mathbf{p} > 0$ (A is symmetric positive definite), and $(\mathbf{p}^T\nabla f(\bar{\mathbf{x}}))^2 \geq 0$. So the answer is yes, the Armijo condition is always true for this α^*.

2. In parts (a)–(d), consider the following constrained minimization problem in $\mathbb{R}^3$:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^3} \quad & -x_1x_2 - x_1 \\ \text{s.t.} \quad & x_1 + x_2 \le 2, \\ & x_1x_3 + x_2x_3 = 1. \end{aligned}$$

(a) **[6 points]** Determine the regularity at every feasible point.

Solution. Case 1. $x_1 + x_2 < 2$. Then there is only 1 active constraint $g(x_1,x_2,x_3) := x_1x_3 + x_2x_3 - 1 = 0$. We check

$$\nabla g(x_1,x_2,x_3) = \begin{pmatrix} x_3 \\ x_3 \\ x_1 + x_2 \end{pmatrix} = \mathbf{0}$$

only when $x_3 = 0$ and $x_1 + x_2 = 0$. But this point violates the constraint $g(x_1,x_2,x_3) = 0$.

Case 2. $x_1 + x_2 = 2$. Then there are 2 active constraints $h(x_1,x_2,x_3) := x_1 + x_2 - 2 = 0$ and $g(x_1,x_2,x_3) = 0$. We have

$$\nabla g(x_1,x_2,x_3) = \begin{pmatrix} x_3 \\ x_3 \\ x_1 + x_2 \end{pmatrix} = \begin{pmatrix} x_3 \\ x_3 \\ 2 \end{pmatrix}, \qquad \nabla h(x_1,x_2,x_3) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

The two gradient vectors are linearly independent since they are not scalar multiples of each other.

Overall, all feasible points are regular.

(b) **[8 points]** Find all the KKT points that satisfy the regularity condition (if any).

Solution. The KKT system is

$$\begin{cases} \begin{pmatrix} -x_2-1 \\ -x_1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} x_3 \\ x_3 \\ x_1+x_2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \mathbf{0}, \\ \mu \geq 0, \\ x_1+x_2 \leq 2, \quad x_1x_3+x_2x_3 = 1, \\ \mu(x_1+x_2-2) = 0. \end{cases}$$

From the third equation, we have $\lambda(x_1+x_2) = 0$ and hence $\lambda = 0$ ($x_1+x_2 \neq 0$, otherwise it violates the equality constraint).

From the first and second equation, we have

$$\mu = x_2+1 = x_1.$$

From the complementary slackness, we have 2 cases:

Case 1.) $\mu = 0$. Then $x_1 = 0$ and $x_2 = -1$. This implies $x_3 = -1$. $(0,-1,-1)$ is a KKT point with multipliers $(0,0)$.

Case 2.) $x_1+x_2 = 2$. Then $x_3 = \frac{1}{2}$. Also, $x_1 = x_2+1$ gives $x_2 = \frac{1}{2}$ and $x_1 = \frac{3}{2}$. $\left(\frac{3}{2},\frac{1}{2},\frac{1}{2}\right)$ is a KKT point with multipliers $(0,\frac{3}{2})$.

Optimization problem in Question 2 for easier reference:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^3} \quad & -x_1x_2 - x_1 \\ \text{s.t.} \quad & x_1 + x_2 \le 2, \\ & x_1x_3 + x_2x_3 = 1. \end{aligned}$$

(c) **[8 points]** For the KKT points found in (b), determine whether it is a local minimizer, local maximizer, or saddle point.

Solution. For $\mathbf{x}^{(1)} = (0,-1,-1)$, $\lambda = 0$, $\mu = 0$: inequality constraint is inactive.

$$H_L(\mathbf{x}^{(1)}) = H_f(\mathbf{x}^{(1)}) = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$Null([\nabla g(\mathbf{x}^{(1)})]^T) = \{(x,y,z) : -x-y-z = 0\}$$

We may take $Z(\mathbf{x}^{(1)}) = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}$. Now $Z(\mathbf{x}^{(1)})^T H_L(\mathbf{x}^{(1)}) Z(\mathbf{x}^{(1)}) = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ is indefinite. Hence $(0,-1,-1)$ is saddle point.

For $\mathbf{x}^{(2)} = (3/2, 1/2, 1/2)$, $\lambda = 0$, $\mu = 3/2$: inequality constraint is active.

$$H_L(\mathbf{x}^{(2)}) = H_f(\mathbf{x}^{(2)}) + 3/2 H_h(\mathbf{x}^{(2)}) = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$Null([\nabla g(\mathbf{x}^{(2)})\ \nabla h(\mathbf{x}^{(2)})]^T) = \{(x,y,z) : 1/2x + 1/2y + 2z = 0,\ x+y = 0\}$$

We may take $Z(\mathbf{x}^{(2)}) = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$. Now $Z(\mathbf{x}^{(2)})^T H_L(\mathbf{x}^{(2)}) Z(\mathbf{x}^{(2)}) = 2 > 0$. Hence $(3/2, 1/2, 1/2)$ is a strict local minimizer.

(d) **[3 points]** Find all global minimizer(s) or explain why it does not exist.

Solution. By considering $x_2 = -2$, $x_3 = \frac{1}{x_1 - 2}$, when $x_1 \to -\infty$ (which is a feasible direction), the objective function value goes to $-\infty$. So global minimizer does not exist.

3. In parts (a)–(d), consider the following constrained minimization problem in $\mathbb{R}^3$:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & x_1^2 + x_2 + 2x_3 \\ \text{s.t.} \quad & x_1 - x_2 + x_3 - 2 \le 0, \\ & \mathbf{x} \in X := \{\mathbf{x} \in \mathbb{R}^3 : x_1 \ge 0,\ x_2 \ge 0,\ x_3 \in \{0,1\}\}. \end{aligned}$$

(a) **[8 points]** Do not modify the set X; keep it as it is. Let $\mu \in \mathbb{R}$ be the Lagrange multiplier associated to the constraint $x_1 - x_2 + x_3 - 2 \le 0$. Write down the Lagrangian function and compute the Lagrangian dual function $\theta(\mu)$.

Solution. The Lagrangian function is

$$L(\mathbf{x},\mu) = x_1^2 + x_2 + 2x_3 + \mu(x_1 - x_2 + x_3 - 2)$$

For any $\mu \ge 0$, the Lagrangian dual function is:

$$\theta(\mu) \quad = \quad \inf_{x_1 \ge 0}\{x_1^2 + \mu x_1\} + \inf_{x_2 \ge 0}\{(1-\mu)x_2\} + \inf_{x_3 \in \{0,1\}}\{(2+\mu)x_3\} - 2\mu$$

For problem 1, since the objective function is convex, the global minimizer is a stationary point satisfying $2x_1 + \mu = 0$ or $x_1 = -\mu/2 \le 0$. Hence for $x_1 \ge 0$, the minimizer is $x_1 = 0$.

$$\begin{aligned} \inf_{x_1 \ge 0}\{x_1^2 + \mu x_1\} &= 0 \\ \inf_{x_2 \ge 0}\{(1-\mu)x_2\} &= \begin{cases} 0, & \text{if } 1-\mu \ge 0, \\ -\infty, & \text{otherwise.} \end{cases} \\ \inf_{x_3 \in \{0,1\}}\{(2+\mu)x_3\} &= \min\{0, 2+\mu\} = 0 \quad \text{since } \mu \ge 0 \end{aligned}$$

So, for $\mu \ge 0$, we have

$$\theta(\mu) = \begin{cases} -2\mu, & \text{if } 0 \le \mu \le 1, \\ -\infty, & \text{otherwise.} \end{cases}$$

(b) **[6 points]** Solve the Lagrangian dual problem.

Solution. The Lagrangian dual problem is

$$\max_{\mu} \ -2\mu, \quad \text{s.t. } 0 \le \mu \le 1.$$

The maximum is attained at $\mu = 0$. The dual optimal value is 0.

(c) **[4 points]** Hence, find an optimal solution to the primal problem via the dual problem. Explain why it is a primal optimal solution.

Solution. We take $\mathbf{x} = (0,0,0)$. Note that $\mathbf{x}$ is feasible and the primal objective value at $\mathbf{x}$ is 0, which is equal to the dual optimal objective value. By the weak duality, $\mathbf{0}$ is a primal optimal solution.

(d) **[4 points]** State the strong duality theorem. Does the strong duality theorem apply to this problem? Justify your answer.

Solution. Strong duality theorem states that for a convex problem satisfying Slater's condition, we have 0 duality gap. However, in this problem, it is not convex (since X is not a convex set). So, strong duality theorem is not applicable.

4. In parts (a) and (b), consider the following constrained minimization problem in $\mathbb{R}^2$:

$$\begin{array}{ll} \min\limits_{x_1,x_2} & x_1 \\ \text{s.t.} & -x_1+x_2-1\le 0 \\ & x_1+x_2-1\le 0, \\ & \mathbf{x}\in X:=\Big\{\mathbf{x}\in\mathbb{R}^2:|x_1|\le 1,\ |x_2|\le 1\Big\}. \end{array}$$

The dual function is given by

$$\theta(\mu_1,\mu_2)=\inf\{x_1+\mu_1(-x_1+x_2-1)+\mu_2(x_1+x_2-1):\mathbf{x}\in X\}.$$

(a) **[6 points]** Determine $\partial\theta(1,1)$, the subdifferential of θ at (1,1). Is this set a singleton?

Solution.

$$\beta(x_1,x_2)=\begin{pmatrix}-x_1+x_2-1\\ x_1+x_2-1\end{pmatrix}$$

To find the set $X(1,1)$, we need to solve the problem:

$$\begin{array}{ll} \min & x_1+2x_2-2 \\ s.t. & |x_1|\le 1,\ |x_2|\le 1. \end{array}$$

The minimizer is at (-1,-1). Hence $\partial\theta(1,1)=\Big\{\big(\beta(-1,-1)\big)\Big\}=\Big\{\begin{pmatrix}-1\\-3\end{pmatrix}\Big\}$. Yes, this set is singleton.

(b) **[6 points]** Determine $\partial\theta(0,0)$, the subdifferential of θ at (0,0). Is this set a singleton?

Solution. To find the set $X(0,0)$, we need to solve the problem:

$$\begin{array}{ll} \min & x_1 \\ s.t. & |x_1|\le 1,\ |x_2|\le 1. \end{array}$$

The minimizer is at $(-1,x_2)$, $-1\le x_2\le 1$. Hence $\partial\theta(0,0)=\Big\{\big(\beta(-1,x_2)\big)\Big\}=\Big\{\begin{pmatrix}x_2\\x_2-2\end{pmatrix},\ -1\le x_2\le 1\Big\}$. No, this set is not singleton.

5. In parts (a)-(d), consider the following equality-constrained convex minimization problem:

$$\text{(ECP)} \quad \begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b}, \end{aligned}$$

where f is convex and differentiable, and $A \in \mathbb{R}^{p\times n}$ satisfies $\text{rank}(A) = p < n$.

(a) **[5 points]** Suppose $f(\mathbf{x}) = x_1^2 + x_1x_2 + x_2^2$, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $b = 1$.
Set $\sigma = 1$, $\lambda^{(0)} = -\frac{3}{2}$. Run one iteration of the augmented Lagrangian method to obtain $\mathbf{x}^{(1)}$.

Solution. The augmented Lagrangian function is

$$\begin{aligned} L_A(\mathbf{x},\lambda,\sigma) &= x_1^2 + x_1x_2 + x_2^2 + \lambda(x_1+x_2-1) + \frac{1}{2\sigma}(x_1+x_2-1)^2 \\ &= x_1^2 + x_1x_2 + x_2^2 - \frac{3}{2}(x_1+x_2-1) + \frac{1}{2}(x_1+x_2-1)^2 \\ &\quad \text{when } \lambda = -3/2,\ \sigma = 1 \end{aligned}$$

$$\nabla L_A(\mathbf{x},\lambda^{(0)},\sigma) = \mathbf{0} \Leftrightarrow \begin{cases} 2x_1 + x_2 - \frac{3}{2} + (x_1+x_2-1) = 0 \\ x_1 + 2x_2 - \frac{3}{2} + (x_1+x_2-1) = 0 \end{cases} \Leftrightarrow \left\{ x_1 = x_2 = 1/2 \right.$$

(b) **[5 points]** Suppose $f(\mathbf{x}) = x_1^2 + x_1x_2 + x_2^2$, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $b = 1$. Set $\mu_0 = 1$. Run one iteration of the quadratic penalty method to obtain $\mathbf{x}^{(1)}$.

Solution. The quadratic penalty function is

$$\begin{aligned} Q(\mathbf{x},\mu) &= x_1^2 + x_1x_2 + x_2^2 + \frac{1}{2\mu}(x_1+x_2-1)^2 \\ &= x_1^2 + x_1x_2 + x_2^2 + \frac{1}{2}(x_1+x_2-1)^2 \quad \text{when } \mu = 1 \end{aligned}$$

$$\nabla Q(\mathbf{x},\mu) = \mathbf{0} \Leftrightarrow \begin{cases} 2x_1 + x_2 + (x_1+x_2-1) = 0 \\ 2x_2 + x_1 + (x_1+x_2-1) = 0 \end{cases} \Leftrightarrow \left\{ x_1 = x_2 = 1/5 \right.$$

(c) **[8 points]** Consider the **projected gradient method** for solving (ECP):

Algorithm 1: Projected Gradient Method for Solving (ECP)

Set $k = 0$. Choose an initial guess $\mathbf{x}^{(0)}$ satisfying $A\mathbf{x}^{(0)} = \mathbf{b}$.
while *stopping criterion not satisfied* **do**
  Compute a direction: $\mathbf{d}^{(k)} = \arg\min_{\mathbf{x}:A\mathbf{x}=\mathbf{0}} \frac{1}{2}\|\mathbf{x} + \nabla f(\mathbf{x}^{(k)})\|^2$
  Use a backtracking line search on f or a fixed step size strategy to find step size α_k.
  Set $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)}$.
  Increment $k \leftarrow k+1$.
end

Explain how the projected gradient method for (ECP) is related to the steepest descent method for the following unconstrained problem:

$$\min_{\mathbf{z}} \; f(F\mathbf{z} + \hat{\mathbf{x}}),$$

where $\hat{\mathbf{x}}$ is any particular solution of $A\mathbf{x} = \mathbf{b}$, and $F \in \mathbb{R}^{n\times(n-p)}$ is any matrix whose range is the nullspace of A satisfying $F^TF = I$.

Solution. For any $\mathbf{x} \in Null(A)$, we can write $\mathbf{x} = F\mathbf{z}$, where F is a matrix whose range is $Null(A)$. Hence, in the projected gradient method, we have

$$\begin{aligned}
\mathbf{d}^{(k)} &= \arg\min_{\mathbf{z}\in\mathbb{R}^n} \frac{1}{2}\|F\mathbf{z} + \nabla f(\mathbf{x}^{(k)})\|^2 \\
&= F\{\mathbf{z} \in \mathbb{R}^n : F^T(F\mathbf{z} + \nabla f(\mathbf{x}^{(k)})) = \mathbf{0}\} \\
&= -FF^T\nabla f(\mathbf{x}^{(k)}) \quad \text{since } F^TF = I
\end{aligned}$$

The critical step in the PGM on (ECP) is then

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha_k FF^T\nabla f(\mathbf{x}^{(k)})$$

For a fix $\hat{\mathbf{x}}$ satisfying $A\mathbf{x} = \mathbf{b}$, the critical step in the steepest descent algorithm for the unconstrained problem

$$\min_{\mathbf{z}} f(F\mathbf{z} + \hat{\mathbf{x}})$$

is

$$\mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} - \alpha_k F^T\nabla f(F\mathbf{z}^{(k)} + \hat{\mathbf{x}}).$$

Hence

$$\begin{aligned}
\mathbf{x}^{(k+1)} &= F\mathbf{z}^{(k+1)} + \hat{\mathbf{x}} \\
&= F\mathbf{z}^{(k)} + \hat{\mathbf{x}} - \alpha_k FF^T\nabla f(F\mathbf{z}^{(k)} + \hat{\mathbf{x}}) \\
&= \mathbf{x}^{(k)} - \alpha_k FF^T\nabla f(\mathbf{x}^{(k)})
\end{aligned}$$

Thus, running the projected gradient from $\mathbf{x}^{(0)}$ is the same as running the steepest descent method on the reduced problem, assuming $F^TF = I$.

(d) **[5 points]** Suppose $f(\mathbf{x}) = x_1^2 + x_1x_2 + x_2^2$, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $b = 1$.
Suppose $\mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Run one iteration of the projected gradient method to obtain $\mathbf{x}^{(1)}$ using fixed step size of $\alpha = 1$.

Solution. Take $F = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. $\nabla f(\mathbf{x}) = \begin{pmatrix} 2x_1 + x_2 \\ 2x_2 + x_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ when $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
Hence,

$$\begin{aligned} \mathbf{x}^{(1)} &= \mathbf{x}^{(0)} - \alpha FF^T \nabla f(\mathbf{x}^{(0)}) \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

(more working space)

(more working space)

(more working space)

End of paper